

AN ANALYTIC MEANS OF DETERMINING THE FLEXURAL AND TORSIONAL RESONANT FREQUENCIES OF GENERALLY ORTHOTROPIC BEAMS

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An analytic method of determining the resonant frequencies of generally orthotropic beams, where flexural to torsional coupling is caused by the character of the elastic material constants, is presented. The analysis predicts that torsional as well as flexural resonant conditions may exist for a freely vibrating beam. A numerical example of the method is presented and compared to experimental results found in the literature. The procedure for determining when a torsional resonant condition exists is explained.

1. INTRODUCTION

The increasing application of advanced composite material systems such as continuous-filament-reinforced composites has generated considerable interest in their analysis. Of particular importance are the unidirectional continuous-filament-reinforced composite material systems, which are highly anisotropic. The analysis and application of such material systems is more challenging than for the more familiar isotropic metals.

The static as well as the dynamic response of unidirectional, continuous-filament-reinforced composite material systems is a function of the direction in which the filaments are oriented within the structural member. A structural member, such as a beam, where the filaments are parallel to the major axis commonly is classified as a transversely isotropic beam [1]. A transversely isotropic beam has only five independent elastic constants, which in terms of the familiar engineering constants are [2] (a list of nomenclature is given in Appendix II)

$$a_{11} = 1/E_{xx}, a_{33} = 1/E_{zz}, a_{12} = -\nu_{yx}/E_{xx}, a_{13} = -\nu_{xz}/E_{zz}, a_{44} = 1/G_{yz}, \quad (1)$$

where E_{ij} are the Young's moduli, ν_{kl} are the Poisson's ratios, and G_{mn} are the shear moduli, in the directions indicated by the subscripts.

If the filaments are oriented at an angle θ to the major axis, z , of the beam while remaining perpendicular to one other of the beam co-ordinate directions, as shown in Figure 1, the beam is classified as being generally orthotropic [1]. Generally orthotropic systems have one plane of elastic symmetry, and thirteen independent elastic constants. For the beam shown in Figure 1, the plane of elastic symmetry is perpendicular to the Y -axis. The elastic constants of a generally orthotropic beam, which may be generated by rotating the filaments of the unidirectional composite material to a given angle, θ , from the major axis, z , can be expressed in terms of the elastic constants of the transversely isotropic material by means of suitable co-ordinate transformations. The principal co-ordinate transformations used in the present study are given in Appendix I.

Generally orthotropic beams possess elastic constants which cause them to exhibit torsional to flexural coupling [1]. That is, if a flexural moment is applied to a generally orthotropic

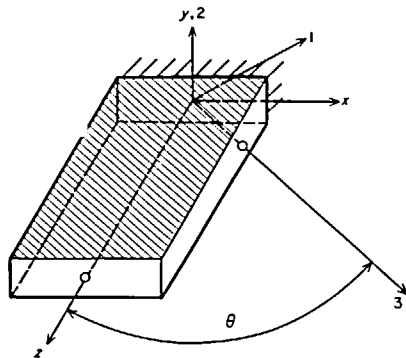


Figure 1. Co-ordinate axes.

beam, the beam will twist about its major axis as well as deform in the plane of the applied moment. A similar phenomenon occurs during the flexural vibrations of a generally orthotropic beam. Thus, a vibrating, generally orthotropic beam exhibits resonant torsional modes as well as flexural modes, when the only forcing function is an externally applied flexural moment. Such torsional resonant modes have been experimentally observed by Abarc ar and Cunniff [3].

If one neglects the non-homogeneity of unidirectional continuous-filament-reinforced composite materials, then the analysis of the response of such materials to a given loading situation, whether static or dynamic, reduces to the analysis of generally orthotropic materials. Unidirectional, continuous-filament-reinforced composites are excellent materials with which to verify the results of a theoretical analysis of generally orthotropic beam behavior because one may obtain a large number of material constants by simply rotating the direction of the filament reinforcement. Although the present analysis for determining the flexural and torsional resonant frequencies of a beam was developed primarily from considerations of beams fabricated of unidirectional, continuous-filament-reinforced materials, it is applicable to any generally-orthotropic beam.

The two best known analyses of the flexural vibrations of beams are the Euler-Bernoulli and the Timoshenko beam theories. The solutions of the Euler-Bernoulli beam theory are well-known, straightforward, and may be found in almost any text dealing with vibrations. However, it also has been shown that the differences between the resonant frequencies predicted by this theory and the experimental data increase with increasing mode number [4]. The difference is caused by the effects of transverse shear stresses and in-plane rotary inertia, which are accounted for in the Timoshenko theory [5]. The effects of transverse shear and in-plane rotary inertia are more pronounced for smaller ratios of the length to the thickness of the beam. However, for beams of typical dimensions, the frequencies predicted by the Euler-Bernoulli beam theory remain comparable to the experimental values even up to the seventh mode and are in fact reasonably accurate for the first few modes. Dudek [4] has used Huang's [6] solutions to the Timoshenko beam theory to show that the more accurate resonant frequency values which are predicted by the Timoshenko theory at higher modes may be used to calculate the material constant G_{yz} for unidirectional, continuous-filament-reinforced composite beams. Dudek has also shown how the effect of transverse shear on the resonant frequencies also increases with larger values of the ratio E_{zz}/G_{yz} , a condition which is prominent for unidirectional composites.

Although the Timoshenko beam theory has been shown to give more accurate values of the flexural resonant frequencies than the Euler-Bernoulli theory, especially for unidirectional composite beams, neither theory is able to predict the torsional resonant frequencies for generally orthotropic beams.

Abarcar and Cunniff [3] have developed a numerical method based on a combination of Myklestad's and Holzer's methods of lumped masses to predict the torsional resonant frequencies of unidirectional composites. Because their method employs massless rods (the fields) between the lumped masses, they were able to include the effects of transverse shear and in-plane rotary inertia. If one tries to extend the equations of motion of the Timoshenko beam theory (which is for a continuous distributed mass beam) to include the torsional displacements of generally orthotropic beams, one obtains three coupled differential equations. The solutions of these three equations however, are, incompatible with the boundary conditions. Therefore, in using a mechanics approach, as in this study, to derive the equations of motion for a generally orthotropic beam, it is necessary to neglect the effects of transverse shear and in-plane rotary inertia.

2. DERIVATION OF THE EQUATIONS OF MOTION

When using a mechanics approach to derive the equations of motion for a beam in either flexure or torsion, one begins by studying a free body diagram of a differential element of the

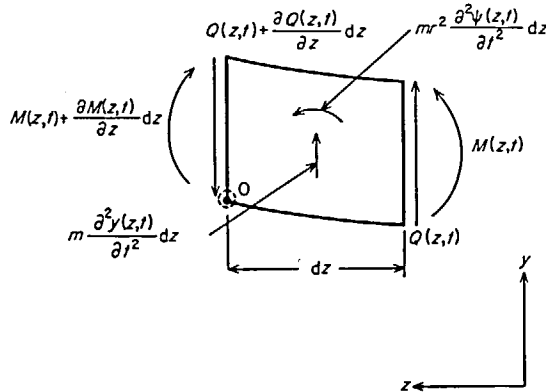


Figure 2. Flexure free body diagram of a differential element of a beam.

beam. From the flexure free body diagram of Figure 2, it can be seen that a summation of flexural moments gives

$$\partial M / \partial z = Q, \quad (2)$$

where M is the flexural moment and Q is the shear force on the cross-section. A summation of vertical forces gives

$$\partial Q / \partial z = w, \quad (3)$$

where w is the distributed load along the beam. For an analysis of free vibrations, the only loads are inertia loads. Therefore the distributed load is

$$w = -m \partial^2 y / \partial t^2 \quad (4)$$

where m is the mass per unit length of the beam and y is the displacement in the corresponding y -co-ordinate direction. Therefore, combining equations (2), (3) and (4), one obtains the equation for flexural motion of a vibrating beam [7]:

$$\frac{\partial^2 M}{\partial z^2} + m \frac{\partial^2 y}{\partial t^2} = 0. \quad (5)$$

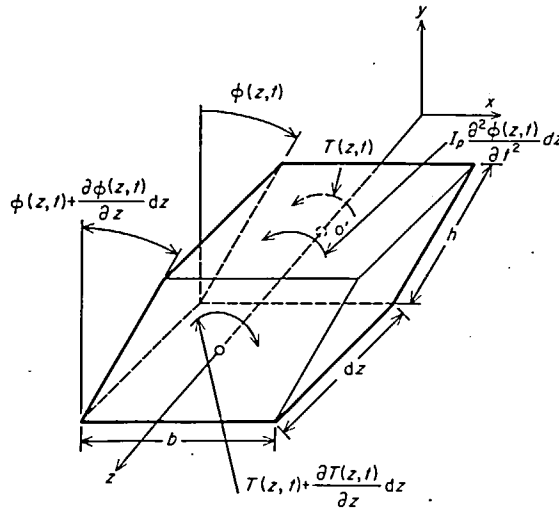


Figure 3. Torsion free body diagram of a differential element of a beam.

Likewise, for torsional vibrations, a mechanics analysis incorporating the torsion free body diagram of Figure 3 yields the following equation representing a summation of torsional moments about the z -axis:

$$\frac{\partial T}{\partial z} - I_p \frac{\partial^2 \phi}{\partial t^2} = 0, \quad (6)$$

where T is the torsional moment about the major beam axis z , ϕ is the torsional displacement, and I_p is the polar mass moment of inertia about the same axis.

Lekhnitskii's results [2] for the static flexural displacement and torsional rotation of a beam having rectilinear anisotropy of a general kind (the most general class of anisotropic material, a generally orthotropic material being a subclass), in which the beam is loaded by a constant flexural moment and a constant torsional moment, are as follows:

$$E_{zz} I_1 \frac{\partial^2 y}{\partial z^2} = M - \frac{E_{zz} I_1}{C_{mt}} T, \quad (7)$$

$$C_t \frac{\partial \phi}{\partial z} = T - \frac{C_t}{C_{mt}} M. \quad (8)$$

The term I_1 is the moment of inertia of the cross-section of the beam about an axis through the axis of the elastic curve and parallel to the x -axis. C_t is the torsional rigidity of the beam. For a generally orthotropic beam, C_t may be approximated as

$$C_t = J G_{xy}, \quad (9)$$

where G_{xy} is the shear modulus in the x - y plane and J is the polar moment of inertia about an axis perpendicular to the cross-section and through the center of twist. The term C_{mt} may be viewed as the torsional to flexural coupling (mutual stiffness) for a beam of generally orthotropic material since

$$C_{mt} = 2I_1/a_{35}. \quad (10)$$

The constant a_{35} is one of the thirteen elastic constants of a generally orthotropic material. The equations relating both G_{xy} and a_{35} to the elastic constants of a transversely isotropic

beam, for a generally orthotropic beam made from a unidirectional composite, are given in Appendix I.

By using equations (7) and (8) to solve for T and M in terms of ϕ and y , one obtains the following:

$$M = \frac{E_{zz} I_1}{\eta} \frac{\partial^2 y}{\partial z^2} + \frac{C_t E_{zz} I_1}{\eta C_{mt}} \frac{\partial \phi}{\partial z}, \quad (11)$$

$$T = \frac{C_t}{\eta} \frac{\partial \phi}{\partial z} + \frac{C_t E_{zz} I_1}{\eta C_{mt}} \frac{\partial^2 y}{\partial z^2}, \quad (12)$$

where η is a dimensionless constant related to the degree of anisotropy of the material. It is defined as

$$\eta = 1 - E_{zz} I_1 C_t / C_{mt}^2 \quad (13)$$

By substituting equations (11) and (12) into equations (5) and (6), respectively, one obtains the two coupled differential equations of motion for a generally orthotropic beam:

$$\frac{E_{zz} I_1}{\eta} \frac{\partial^4 y}{\partial z^4} + \frac{C_t E_{zz} I_1}{\eta C_{mt}} \frac{\partial^3 \phi}{\partial z^3} + m \frac{\partial^2 y}{\partial t^2} = 0, \quad (14)$$

$$\frac{C_t}{\eta} \frac{\partial^2 \phi}{\partial z^2} + \frac{C_t E_{zz} I_1}{\eta C_{mt}} \frac{\partial^3 y}{\partial z^3} - I_p \frac{\partial^2 \phi}{\partial t^2} = 0. \quad (15)$$

It is important to note that for a transversely isotropic or an isotropic beam, there exists no coupling by the elastic constants between the flexural and torsional displacements: that is, $a_{35} = 0$; thus $1/C_{mt} = 0$ and $\eta = 1$. For this case, the differential equations of motion (equations (14) and (15)) revert to the classical forms describing the flexural vibration and torsional vibration of isotropic and transversely isotropic beams.

Manipulating equations (14) and (15) to eliminate either of the terms y or ϕ results in two separate (decoupled) sixth-order, homogeneous, partial differential equations with constant coefficients, each in terms of a single dependent variable. The equation in terms of y is as follows:

$$\frac{\partial^6 y}{\partial z^6} - \frac{I_p}{C_t} \frac{\partial^6 y}{\partial z^4 \partial t^2} + \frac{m}{E_{zz} I_1} \frac{\partial^4 y}{\partial z^2 \partial t^2} - \frac{\eta I_p m}{C_t E_{zz} I_1} \frac{\partial^4 y}{\partial t^4} = 0. \quad (16)$$

An equation identical to equation (16) in terms of ϕ also exists. It may be obtained by replacing y with ϕ .

The solutions to these two equations, one for ϕ and one for y , then will describe the flexural and torsional displacements for a generally orthotropic beam. All that remains is to find the solutions and apply the proper boundary conditions.

3. SOLUTIONS OF THE DIFFERENTIAL EQUATIONS OF MOTION

Since the coupled differential equations of motion, equations (14) and (15), are postulated to describe the free vibration of a beam, it is logical to assume that the solutions to the resulting sixth-order, decoupled partial differential equations will be harmonic with respect to time, t , having circular frequency p . It also will be assumed that the solutions for y and ϕ have independent normal functions in terms of z . That is, solutions of the following forms will be assumed:

$$y(z, t) = Y(z) \exp(ipt), \quad (17)$$

$$\phi(z, t) = \Phi(z) \exp(ipt), \quad (18)$$

where $Y(z)$ and $\Phi(z)$ are the yet undetermined normal functions of the solutions.

By non-dimensionalizing the normal functions and substituting the assumed solutions into the decoupled partial differential equations, such as the one given by equation (16), one obtains decoupled total differential equations. The resulting total differential equation for Y is as follows:

$$Y^{VI} + \frac{p^2 L^2 I_p}{C_t} Y^{IV} - \frac{p^2 L^4 m}{E_{zz} I_1} Y^{II} - \frac{\eta p^4 L^6 m I_p}{C_t E_{zz} I_1} Y = 0. \quad (19)$$

An identical equation to equation (19) exists for Φ , where $Y = Y(\xi)$, and $\Phi = \Phi(\xi)$. The term ξ is the non-dimensional length: i.e., $\xi = z/L$, where L is the length of the beam. The superscripts on Y indicate differentiation with respect to ξ : for example,

$$Y^{VI} = d^6 Y/d\xi^6.$$

The assumed solutions (equations (17) and (18)) also must be solutions to the coupled equations (14) and (15). Therefore, by substituting the assumed solutions into equations (14) and (15) and performing the differentiation with respect to time, the coupled equations can be reduced to the following total differential equations in terms of ξ :

$$Y^{IV} + \frac{C_t L}{C_{mt}} \Phi^{III} - \frac{\eta m p^2 L^4}{E_{zz} I_1} Y = 0, \quad (20)$$

$$\Phi^{II} + \frac{E_{zz} I_1}{C_{mt} L} Y^{III} + \frac{\eta I_p p^2 L^2}{C_t} \Phi = 0. \quad (21)$$

The normal functions Y and Φ which satisfy the respective sixth-order total differential equations, such as equation (19), are assumed to be of the following forms:

$$Y(\xi) = A \exp(\omega \xi), \quad (22)$$

$$\Phi(\xi) = A' \exp(\omega' \xi), \quad (23)$$

where ω and ω' are solution roots of the normal functions. Upon substituting these expressions into the decoupled total differential equations, such as equation (19), characteristic equations for ω and ω' are found. The characteristic equation for ω is as follows:

$$\omega^6 + \frac{p^2 I_p L^2}{C_t} \omega^4 - \frac{p^2 L^4 m}{E_{zz} I_1} \omega^2 - \frac{p^4 m \eta I_p L^6}{C_t E_{zz} I_1} = 0, \quad (24)$$

and an identical equation exists for ω' .

It is evident that, for a given beam with its associated parameters, the solution set for ω equals the solution set for ω' . Therefore, the normal function $\Phi(\xi)$ may be expressed as

$$\Phi(\xi) = A' \exp(\omega \xi). \quad (25)$$

In order to obtain the roots of the characteristic equation (24), the following substitutions will be made:

$$\lambda = \omega^2, \quad A = I_p/C_t, \quad B = p^2 L^2, \quad C = m/E_{zz} I_1. \quad (26)$$

The following cubic equation results:

$$\lambda^3 + AB\lambda^2 - BCL^2\lambda - AB^2C\eta L^2 = 0. \quad (27)$$

In general there exist three possible branches to the roots of equation (27), depending on the values of the coefficients of λ [8]. The significance of each branch is as follows.

3.1. BRANCH 1: THREE REAL AND UNEQUAL ROOTS

For this case it is necessary that

$$\frac{(2A^3 B^3 + 9AB^2 CL^2 - 27AB^2 C\eta L^2)^2}{2916} + \frac{(-3BCL^2 - A^2 B^2)^3}{729} < 0. \quad (28)$$

This is the predominate branch for many generally orthotropic beams. The three roots are real and unequal; however, each may be either positive or negative. If a root λ_i is negative, then the two roots ω_{i1} and ω_{i2} of the characteristic equation (24), corresponding to λ_i , are complex conjugates. For a positive root λ_i , the two corresponding roots of equation (24) are real, one being positive and one negative. There exist four possible combinations of signs of the three roots of λ_i .

- (i) *All three roots positive.* For this combination, the six roots of equation (24) are all real; three are positive and three are negative.
- (ii) *Two roots positive; one negative.* Here, four of the roots of equation (24) are real, and two are complex conjugates.
- (iii) *One root positive; two negative.* This combination yields two real roots of equation (24), and two sets of complex conjugates.
- (iv) *All three roots negative.* Three sets of complex conjugate roots exist for equation (24).

The normal functions, for these four cases of all real roots to equation (27), are of the forms:

$$Y(\xi) = A_1 \exp(\sqrt{\lambda_1} \xi) + A_2 \exp(-\sqrt{\lambda_1} \xi) + A_3 \exp(\sqrt{\lambda_2} \xi) + A_4 \exp(-\sqrt{\lambda_2} \xi) + A_5 \exp(\sqrt{\lambda_3} \xi) + A_6 \exp(-\sqrt{\lambda_3} \xi), \quad (29)$$

$$\Phi(\xi) = A'_1 \exp(\sqrt{\lambda_1} \xi) + A'_2 \exp(-\sqrt{\lambda_1} \xi) + A'_3 \exp(\sqrt{\lambda_2} \xi) + A'_4 \exp(-\sqrt{\lambda_2} \xi) + A'_5 \exp(\sqrt{\lambda_3} \xi) + A'_6 \exp(-\sqrt{\lambda_3} \xi), \quad (30)$$

where A_i and A'_i are two sets of integration constants containing six elements each. These two sets of constants are related to each other since the normal functions of equations (29) and (30) also must satisfy the coupled differential equations (20) and (21). The set of constants A'_i can be expressed in terms of the set A_i . The forms of the six algebraic equations describing the relationships between the two sets are dependent on which combination of roots λ_i result. For most beam configurations and elastic material parameters of practical interest, the combination of roots discussed under case (iii) is the most likely to occur. An example of the forms of the six algebraic equations relating the set of constants A'_i to the set A_i for this combination of roots will be given in section 5.

3.2. BRANCH 2: ONE REAL ROOT AND ONE SET OF COMPLEX CONJUGATE ROOTS

For this branch it is necessary that

$$\frac{(2A^3 B^3 + 9AB^2 CL^2 - 27AB^2 C\eta L^2)^2}{2916} + \frac{(-3BCL^2 - A^2 B^2)^3}{729} > 0. \quad (31)$$

This branch may occur at higher modes of vibration when the combination of elastic material constants and beam configurations yields a negative value of η : i.e., when $E_{zz} I_1 C_t / C_{mt}^2 > 1$. For this branch there exists only one set of solution roots ω_i to equation (24). Upon assuming

that λ_1 represents the real root, then λ_2 and λ_3 are complex conjugates. With the definitions

$$\alpha = \left[\frac{\sqrt{G^2 + 4H^3} - G}{2} \right]^{1/3}, \quad (32)$$

$$G = \frac{2A^3 B^3 + 9AB^2 CL^2 + 27AB^2 C\eta L^2}{27}, \quad (33)$$

$$H = \frac{-3BCL^2 - A^2 B^2}{9}, \quad (34)$$

the solutions for λ_i become,

$$\begin{aligned} \lambda_1 &= -\frac{AB}{3} + \left(\alpha - \frac{H}{\alpha} \right), \\ \lambda_2 &= -\frac{AB}{3} - \frac{1}{2} \left(\alpha - \frac{H}{\alpha} \right) + \frac{\sqrt{3}}{2} \left(\alpha + \frac{H}{\alpha} \right) i, \\ \lambda_3 &= -\frac{AB}{3} - \frac{1}{2} \left(\alpha - \frac{H}{\alpha} \right) - \frac{\sqrt{3}}{2} \left(\alpha + \frac{H}{\alpha} \right) i, \end{aligned} \quad (35)$$

where $i = \sqrt{-1}$. The roots of equation (24) are then the square roots of the solutions given in equation (35). With the definitions

$$u = - \left[\frac{AB}{3} + \frac{1}{2} \left(\alpha - \frac{H}{\alpha} \right) \right], \quad (36)$$

$$v = \frac{\sqrt{3}}{2} \left(\alpha + \frac{H}{\alpha} \right), \quad (37)$$

$$\gamma = \arctan \frac{v}{u}, \quad (38)$$

the roots ω_i become

$$\omega_1 = +\sqrt{\lambda_1}, \quad (39)$$

$$\omega_2 = -\sqrt{\lambda_1}, \quad (40)$$

$$\omega_3 = (u^2 + v^2)^{1/4} \cos \frac{\gamma}{2} + i(u^2 + v^2)^{1/4} \sin \frac{\gamma}{2}, \quad (41)$$

$$\omega_4 = -(u^2 + v^2)^{1/4} \cos \frac{\gamma}{2} - i(u^2 + v^2)^{1/4} \sin \frac{\gamma}{2}, \quad (42)$$

$$\omega_5 = -(u^2 + v^2)^{1/4} \cos \frac{\gamma}{2} + i(u^2 + v^2)^{1/4} \sin \frac{\gamma}{2}, \quad (43)$$

$$\omega_6 = (u^2 + v^2)^{1/4} \cos \frac{\gamma}{2} - i(u^2 + v^2)^{1/4} \sin \frac{\gamma}{2}. \quad (44)$$

The normal functions are then

$$Y(\xi) = A_1 \exp(\sqrt{\lambda_1} \xi) + A_2 \exp(-\sqrt{\lambda_1} \xi) + A_3 \exp(\Gamma + i\beta) + \\ + A_4 \exp(-\Gamma - i\beta) + A_5 \exp(-\Gamma + i\beta) + A_6 \exp(\Gamma - i\beta), \quad (45)$$

$$\Phi(\xi) = A'_1 \exp(\sqrt{\lambda_1} \xi) + A'_2 \exp(-\sqrt{\lambda_1} \xi) + A'_3 \exp(\Gamma + i\beta) + \\ A'_4 \exp(-\Gamma - i\beta) + A'_5 \exp(-\Gamma + i\beta) + A'_6 \exp(\Gamma - i\beta), \quad (46)$$

where

$$\Gamma = (u^2 + v^2)^{1/4} \cos \frac{\gamma}{2} \quad (47)$$

$$\beta = (u^2 + v^2)^{1/4} \sin \frac{\gamma}{2}. \quad (48)$$

Since the set of constants A_i and A'_i are as of yet arbitrary, the normal functions $Y(\xi)$ and $\Phi(\xi)$ (equations (45) and (46)) may be written in the following forms:

$$Y(\xi) = A_1 \cos(\sqrt{\lambda_1} \xi) + A_2 \sin(\sqrt{\lambda_1} \xi) + A_3 \cos \beta \xi \cosh \Gamma \xi + \\ + A_4 \cos \beta \xi \sinh \Gamma \xi + A_5 \sin \beta \xi \cosh \Gamma \xi + A_6 \sin \beta \xi \sinh \Gamma \xi, \quad (45a)$$

$$\Phi(\xi) = A'_1 \sin(\sqrt{\lambda_1} \xi) + A'_2 \cos(\sqrt{\lambda_1} \xi) + A'_3 \cos \beta \xi \sinh \Gamma \xi + \\ + A'_4 \cos \beta \xi \cosh \Gamma \xi + A'_5 \sin \beta \xi \sinh \Gamma \xi + A'_6 \sin \beta \xi \cosh \Gamma \xi. \quad (46a)$$

The set of constants A'_i can be expressed in terms of the set of constants A_i (since there exists only one combination of roots for this branch), by substituting the normal functions of equations (45a) and (46a) into either of the coupled differential equations (20) or (21). These relationships are as follows:

$$A'_1 = (C_{mt}/C_t L \lambda_1^{3/2}) (\lambda_1^2 - \eta CBL^2) A_1, \quad (49)$$

$$A'_2 = -(C_{mt}/C_t L \lambda_1^{3/2}) (\lambda_1^2 - \eta CBL^2) A_2, \quad (50)$$

$$A'_3 = (C_{mt}/C_t L) \left[\frac{e}{a} A_6 - \frac{d}{a} A_3 \right], \quad (51)$$

$$A'_4 = (C_{mt}/C_t L) \left[\frac{e}{a} A_5 - \frac{d}{a} A_4 \right], \quad (52)$$

$$A'_5 = (C_{mt}/C_t L) \left[-\frac{e}{a} A_4 - \frac{d}{a} A_5 \right], \quad (53)$$

$$A'_6 = (C_{mt}/C_t L) \left[-\frac{e}{a} A_3 - \frac{d}{a} A_6 \right], \quad (54)$$

where

$$a = \Gamma^2(\Gamma^2 - 3\beta^2)^2 + \beta^2(\beta^2 - 3\Gamma^2)^2, \quad (55)$$

$$d = \Gamma(\Gamma^2 - 3\beta^2)(\Gamma^4 - 6\Gamma^2\beta^2 + \beta^4 - \eta CBL^2) + 4\Gamma\beta^2(\beta^2 - \Gamma^2)(\beta^2 - 3\Gamma^2). \quad (56)$$

$$e = 4\Gamma^2\beta(\beta^2 - \Gamma^2)(\Gamma^2 - 3\beta^2) - \beta(\beta^2 - 3\Gamma^2)(\Gamma^4 - 6\Gamma^2\beta^2 + \beta^4 - \eta CBL^2). \quad (57)$$

3.3. BRANCH 3: THREE REAL ROOTS; AT LEAST TWO EQUAL

Although this is a possible branch for the roots of the cubic equation (27), it is rather redundant when considerations of the physical parameters involved in vibrating generally orthotropic beams are taken into account. In order to have this branch, it is necessary that

$$\frac{(2A^3 B^3 + 9AB^2 CL^2 - 27AB^2 C\eta L^2)^2}{2916} + \frac{(-3BCL^2 - A^2 B^2)^3}{729} = 0. \quad (58)$$

By substituting the definitions of A , B and C in terms of the beam parameters (equations (26)) into equations (58) and performing some rather tedious algebraic manipulations, one arrives at the following form:

$$\eta p^4 + (mC_t^2/4I_p^2 E_{zz} I_1)(1 + 18\eta - 27\eta^2)p^2 - (m^2 C_t^4/E_{zz}^2 I_1^2 I_p^4) = 0. \quad (59)$$

Thus, this branch may occur only if the frequency of the beam has the following value:

$$p = \sqrt{(mC_t^2/8\eta I_p^2 E_{zz} I_1)[(27\eta^2 - 18\eta - 1) \pm \sqrt{(27\eta^2 - 18\eta - 1)^2 + 64\eta}]} \quad (60)$$

The right-hand side of equation (60) is a function only of the cross-sectional shape of the beam and the elastic material constants.

In order for a beam to be vibrating at a resonant frequency where the normal functions of the displacements are given by the roots of this branch, it is necessary that the resonant frequency be equal to the value given by equation (60). However, the frequency value given by equation (60), in general, will not be a resonant frequency. Since the frequency values given by equation (60) are dependent only on the cross-sectional shape and elastic materials constants, and not on the boundary conditions or the length of the beam, it would be only coincidental that this frequency would match a resonant frequency value.

The possibility that this branch ever can be valid is decreased further by the fact that, for some combinations of beam parameters, this branch never can be attained. For example, when the value of η is negative there exist no real frequencies which satisfy equation (59). The proof lies in equation (60). If η is within the range $0 \geq \eta \geq -0.19$, the argument of the inner radical of equation (60) is negative. Thus, the frequency must be imaginary in order to satisfy equation (59). This is physically impossible. For the range $-0.19 > \eta > -\infty$, the value of the quantity within the inner bracket is positive; thus the argument of the outer radical is negative and again the frequency is imaginary. If η is equal to zero, the branch is invalid, because the frequency value of equation (60) is then infinite.

When the value of η is positive, it is possible to obtain a real positive frequency value from equation (60). However, only one real positive frequency exists for positive values of η . The absolute value of the inner radical is always greater than the value of the expression within the parentheses, for positive η . Therefore, the expression within the brackets will have one positive and one negative value. The sign of the argument of the outer radical follows the sign of the expression within the inner brackets. Thus, the frequency will be real only for the positive value of the expression within the brackets, and only the positive value of the frequency has physical significance.

Therefore, for positive values of η , one must calculate the value given in equation (60), to determine that the resonant frequencies of interest are not near this value. This procedure, in most cases, is hardly necessary since for nominal beam dimensions and realistic elastic material constants, this frequency value (of the order of 1×10^6 to 1×10^{10} Hertz) is well beyond the normal range of interest.

4. BOUNDARY CONDITIONS

The equations of motion and the general forms of the normal functions which were presented in the previous two sections hold true for all generally orthotropic vibrating beams. The determination of the set of constants of integration, A_i or A'_i , and the resonant frequencies are dependent upon the type of beam supports. Six boundary conditions are necessary in order to determine these six constants of integration. (It is noted that there exist only six integration constants to be determined, and not twelve, since the set of constants A'_i have been shown to be dependent upon A_i).

Three types of end supports commonly are discussed when referring to the analysis of a vibrating beam. They are defined as the free end, the hinged end, and the clamped end. The three boundary conditions associated with each type of support are defined as follows:

(1) hinged end;

$$Y(\xi) = 0, \quad (61)$$

$$\Phi(\xi) = 0, \quad (62)$$

$$Y'''(\xi) = 0: \quad (63)$$

(2) free end;

$$Y''(\xi) = 0, \quad (64)$$

$$Y'''(\xi) = 0, \quad (65)$$

$$\Phi'(\xi) = 0: \quad (66)$$

(3) clamped end;

$$Y(\xi) = 0, \quad (67)$$

$$Y'(\xi) = 0, \quad (68)$$

$$\Phi(\xi) = 0. \quad (69)$$

It is important to note that all of the boundary conditions are homogeneous and independent.

If one substitutes the solution forms of the normal functions, either equations (29) and (30) or equations (45a) and (46a), depending upon which solution branch is involved, into the boundary conditions representing the desired support conditions, six homogeneous algebraic equations in terms of either the set of constants A_i or A'_i are obtained. These six equations may be written in matrix form. The resulting matrix equation is also homogeneous; hence non-trivial solutions occur only when the determinant of the coefficients of the set of constants A_i or A'_i is equal to zero. The determinant of the coefficient matrix thus yields the characteristic matrix from which the eigenfrequencies, i.e., the resonant frequencies, of the beam may be determined.

The determinant of this coefficient matrix for a generally orthotropic beam must be formed from a sixth order square matrix. The algebraic manipulation involved to obtain the characteristic equation and the subsequent determination of the roots of the equation become rather unwieldy. However, since the coefficients are functions of the frequency, p , it is a much more convenient process to simply iterate through a range of frequencies and evaluate the determinant. The frequency at which the value of the determinant vanishes is then a resonant frequency of the beam. A demonstration of this procedure, as an application of the previously discussed theory, will be presented in the following example involving unidirectional, continuous-filament-reinforced graphite/epoxy composite materials cantilever beams.

5. NUMERICAL EXAMPLE

The vibrational modes of unidirectional, continuous-filament-reinforced graphite/epoxy and continuous-filament-reinforced boron/epoxy cantilever beams were studied by Abarcar and Cuniff [3]. In their study, the filaments were aligned at either zero, fifteen, or thirty degrees to the z -axis of the beam (see Figure 1). The beams in which the filaments were aligned at either fifteen or thirty degrees to the z -axis therefore were generally orthotropic. Resonant torsional as well as flexural vibrational modes were observed for all of the generally orthotropic beams studied.

The present theory was applied to all of the beam configurations tested by Abarcar and Cuniff [3]. The resonant frequencies for the first few flexural and the first few torsional modes were calculated and, in most cases, they agreed fairly well with the experimental values. For brevity, only the results of a continuous-filament graphite/epoxy beam with the filaments aligned at fifteen degrees will be presented here, to demonstrate the procedure.

The beams under consideration are cantilever beams: i.e., clamped at one end and free at the other. The nominal dimensions of the beam under consideration are L = length = 19.05 cm (7.5 inches), b = width = 1.27 cm (0.50 inches) and h = thickness = 0.317 cm (0.125 inches). The density of the material is $\rho = 48.18 \text{ kg/m}^3$ (0.056 lb/in³). The important elastic material constants are $E_{33} = 122.10 \text{ GPa}$ ($18.71 \times 10^6 \text{ lb/in}^2$), $E_{11} = 9.39 \text{ GPa}$ ($1.362 \times 10^6 \text{ lb/in}^2$) and $\nu_{13} = 0.30$. Abarcar and Cuniff, in effect, listed three values of the shear modulus G_{13} [3]. They calculated the value of G_{13} from a zero degree beam using Dudek's approach [4]. They also listed two other values of G_{13} , one calculated from their data for a thirty degree beam, and the other from their data for a fifteen degree beam.

Since the solution for the resonant frequencies is rather sensitive to the assumed value of G_{13} , results for all three values are presented in Table 1. The three values of the shear modulus are $G_{13} = 4.30 \text{ GPa}$ ($0.6242 \times 10^6 \text{ lb/in}^2$), from a zero degree beam, $G_{13} = 5.84 \text{ GPa}$ ($0.8465 \times 10^6 \text{ lb/in}^2$), from a fifteen degree beam, and $G_{13} = 5.16 \text{ GPa}$ ($0.7479 \times 10^6 \text{ lb/in}^2$), from a thirty degree beam. For each assumed value of G_{13} , the beam elastic constants have been calculated by using the relations of Appendix I, for $\theta = 15^\circ$. With these values at hand, an iteration through a range of frequencies is then performed. Because of the relatively large number of calculations involved in the iteration process, computer assistance is convenient.

The iteration process is performed as follows. First, for an assumed frequency, the values of the constants in equation (26) are calculated. Then a check is made to determine which of the conditions given by equations (28), (31) or (58) is satisfied: i.e., to determine which branch of the roots of the cubic equation (27), is applicable in computing the normal functions of the beam. The first branch, i.e., three separate real roots, was applicable for all the frequencies obtained in the example of Table 1. One root of this branch was always positive, with two negative roots: i.e., combination number (iii) of Branch 1 applied for all frequencies considered in conjunction with this example beam. Since the two sets of integration constants A_i and A'_i are still arbitrary at this point, the normal functions (equations (29) and (30) for this combination of roots, λ_i , may be written as

$$Y(\xi) = A_1 \cosh(\sigma_1 \xi) + A_2 \sinh(\sigma_1 \xi) + A_3 \cos(\sigma_2 \xi) + A_4 \sin(\sigma_2 \xi) + A_5 \cos(\sigma_3 \xi) + A_6 \sin(\sigma_3 \xi), \quad (29a)$$

$$\Phi(\xi) = A'_1 \sinh(\sigma_1 \xi) + A'_2 \cosh(\sigma_1 \xi) + A'_3 \sin(\sigma_2 \xi) + A'_4 \cos(\sigma_2 \xi) + A'_5 \sin(\sigma_3 \xi) + A'_6 \cos(\sigma_3 \xi), \quad (30a)$$

where

$$\sigma_i = \sqrt{|\lambda_i|}. \quad (70)$$

TABLE 1
Comparison of resonant frequency calculations for a fifteen degree
graphite/epoxy beam

Mode number	Experimental values from reference [3] (Hz)	Calculated values from the present analysis			Calculated values from the Euler-Bernoulli theory (Hz)
		$G_{13} = 5.84 \text{ GPa}$ ($0.8465 \times 10^6 \text{ lb/in}^2$) (Hz)	$G_{13} = 5.16 \text{ GPa}$ ($0.7479 \times 10^6 \text{ lb/in}^2$) (Hz)	$G_{13} = 4.30 \text{ GPa}$ ($0.6424 \times 10^6 \text{ lb/in}^2$) (Hz)	
1	83	84	81	77	84
2	511	532	513	490	527
3	1423	1320	1062†	880†	1475
	1527†	1507†	1458	1397	—
4	2784	2974	2892	2780	2891
5	4365	4980	4578†	3800†	4778
	4732†	5677†	4844	4687	—
6	—	—	6921	6942	7139

† Indicates torsional modes.

TABLE 2
Matrix of boundary conditions

	1	0	1	0	1	0		A ₁
1								0
0		$\psi_1 \delta_1$	0	$\psi_2 \delta_2$	0		$\psi_3 \delta_3$	A ₂
0		σ_1	0	σ_2	0		σ_3	A ₃
$\sigma_1^2 \cosh \sigma_1$		$\sigma_1^2 \sinh \sigma_1$	$-\sigma_2^2 \cos \sigma_2$	$-\sigma_2^2 \sin \sigma_2$	$-\sigma_3^2 \cos \sigma_3$		$-\sigma_3^2 \sin \sigma_3$	A ₄
$\sigma_1^3 \sinh \sigma_1$		$\sigma_1^3 \cosh \sigma_1$	$\sigma_2^2 \sin \sigma_2$	$-\sigma_2^3 \cos \sigma_2$	$\sigma_3^3 \sin \sigma_3$		$-\sigma_3^3 \cos \sigma_3$	A ₅
$\sigma_1 \psi_1 \delta_1 \cosh \sigma_1$		$\sigma_1 \psi_1 \delta_1 \sinh \sigma_1$	$-\sigma_2 \psi_2 \delta_2 \cos \sigma_2$	$-\sigma_2 \psi_2 \delta_2 \sin \sigma_2$	$\sigma_3 \psi_3 \delta_3 \cos \sigma_3$		$-\sigma_3 \psi_3 \delta_3 \sin \sigma_3$	A ₆

= [0]

With functions given by equations (29a) and (30a), the set of constants A'_i , expressed in terms of A_i , are

$$\begin{aligned} A'_1 &= \psi_1 \delta_1 A_1, A'_2 = \psi_1 \delta_1 A_2, A'_3 = -\psi_2 \delta_2 A_3, \\ A'_4 &= \psi_2 \delta_2 A_4, A'_5 = -\psi_3 \delta_3 A_5, A'_6 = \psi_3 \delta_3 A_6, \end{aligned}$$

where

$$\psi_i = (\eta m p^2 L^4 / E_{zz} I_1) - \sigma_i^4, \delta_i = C_{mi} / C_i L \sigma_i^3.$$

By substituting the normal functions of equations (29a) and (30a) into the boundary conditions, and then expressing each constant A'_i in terms of its complement A_i , the homogeneous matrix equation shown in Table 2 is obtained. It is the determinant of the coefficient matrix of the vector $\mathbf{A}$ which must vanish as a resonant frequency is approached. Thus, the lowest frequency at which the determinant is zero is the first mode, etc. However, the determinant will vanish when the frequency reaches a resonant torsional mode as well as a flexural mode. One then must determine which of the resonant frequencies correspond to a torsional mode, and which correspond to a flexural mode.

In order to distinguish between a torsional mode and a flexural mode, it is only necessary to calculate the resonant frequencies by using either the Timoshenko or the Euler-Bernoulli beam theory, with the calculated values of Young's modulus in the z -direction, E_{zz} , and the calculated shear modulus, G_{xy} , of the generally orthotropic beam. The resonant frequencies which these theories provide are necessarily flexural modes only. The difference in the values of the flexural resonant frequencies determined by using either of these analyses and the present analysis, are small enough to allow direct comparison, for the first few modes at least. Therefore, by comparing a list of frequencies determined from the present theory to the first few resonant frequencies provided by the Euler-Bernoulli theory (the Euler-Bernoulli frequencies are easier to calculate than the ones from the Timoshenko theory), the resonant frequencies corresponding to a torsional mode can be identified: i.e., there will be no corresponding frequency for a torsional mode in the Euler-Bernoulli set. For comparison purposes, the resonant frequencies as calculated from the Euler-Bernoulli theory are listed in the last column of Table 1.

6. DISCUSSION AND CONCLUSIONS

The results presented in Table 1 illustrate the strong dependence of the resonant frequencies of a generally orthotropic beam upon the assumed value of the shear modulus, G_{13} . It is interesting to note that for rather small changes in these shear moduli, the resonant frequencies of the torsional modes change much more than the resonant frequencies for the flexural modes. The present analysis, in fact, predicts that the order of occurrence of torsional and flexural resonance at a given mode may change: e.g., the torsional modes are predicted to occur before the corresponding flexural modes when $G_{13} = 5.16$ GPa (0.7479×10^6 lb/in²), and after the flexural modes when $G_{13} = 5.84$ GPa (0.8465×10^6 lb/in²), as shown in Table 1. Such a phenomenon is physically quite realistic.

The normal functions $Y(\xi)$ and $\Phi(\xi)$ also can be determined for each resonant condition if so desired. This entails determining the values of the two sets of constants A_i and A'_i for a given resonant frequency. It is thus possible to determine the mode shapes of the beam for both the lateral displacement and the torsional displacement.

The present analysis is indicated to be quite adequate for determining the resonant frequencies of generally orthotropic beams and their corresponding mode shapes. The accuracy of the predicted resonant frequencies of the cantilever beam being discussed here is just as

good as for the values predicted by numerical analysis methods [3], as determined by comparing both to the experimentally measured values.

It is realized that the present analysis is still not a completely accurate model of the physical material; however, an even more accurate determination of the resonant frequencies would be possible if a more refined analysis, which accounts for all of the elastic constants of a generally orthotropic material, were devised. Such a refined analysis will undoubtedly be complicated to formulate, however, and its application probably would be hindered by the complexity involved.

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APPENDIX I

The following are the principal generally orthotropic beam elastic material constants used in the present analysis; they are obtained by co-ordinate rotations, the material elastic constants of a transversely isotropic material being used:

$$a_{33} = 1/E_{zz} = \frac{\sin^4 \theta}{E_{11}} + \frac{\cos^4 \theta}{E_{33}} + \left(\frac{1}{G_{13}} - \frac{2\nu_{13}}{E_{33}} \right) \cos^2 \theta \sin^2 \theta,$$

$$a_{66} = 1/G_{xy} = \frac{\sin^2 \theta}{G_{13}} + \frac{\cos^2 \theta}{G_{12}},$$

$$a_{35} = 2 \left(\frac{\sin^2 \theta}{E_{11}} - \frac{\cos^2 \theta}{E_{33}} \right) \sin \theta \cos \theta + \left(\frac{1}{G_{13}} - \frac{2\nu_{13}}{E_{33}} \right) (\cos^2 \theta - \sin^2 \theta) \sin \theta \cos \theta$$

APPENDIX II: NOMENCLATURE

- a_{ij} material elastic compliance values
- b beam width
- $a, d, e, \psi_i, \delta_i$ constant parameters
- h beam thickness
- m mass per unit length of beam
- p circular frequency
- t time

w	load per unit length along beam
x	distance along x -axis
y	distance along y -axis
z	distance along z -axis
A_i, A'_i	integration constants
A, B, C	parameters of cubic characteristic equation
C_t	torsional rigidity
C_{mt}	mutual stiffness coefficient
E_{ii}	Young's moduli
G_{ij}	shear moduli
I_x	moment of inertia about x -axis
I_p	polar mass moment of inertia about z -axis
J	polar moment of inertia about z -axis
L	beam length
M	flexural moment
Q	shear force on a cross-section
T	torsional moment about z -axis
Γ, β	dimensionless parameters of sixth-order equations
λ_i	roots of cubic characteristic equation
η	dimensionless constant relating degree of anisotropy of the material
ν_{ij}	Poisson's ratios
ξ	non-dimensional length
ρ	material density
σ_i	square root of the absolute value of λ_i
ϕ	torsional displacement
ω_i, ω'_i	solution roots to the normal functions
Y	normal function of lateral displacement (a function of ξ)
Φ	normal function of torsional displacement (a function of ξ)